DATASET PROFILING & SUMMARY STATS REPORT

Dataset Used: Adult Psychiatric Morbidity Survey (APMS) 2014

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1. Executive Summary:

This report details the structural audit of the Adult Psychiatric Morbidity Survey (APMS) series, including the 2014 core chapters and 2024 updates. We have successfully programmatically inventoried 181 tables, establishing a high-fidelity baseline for clinical prevalence, treatment gaps, and demographic risk factors.

2. Technology & Tools Stack:

To execute this programmatic audit and structural profiling, the following technical environment was utilized:

- **Language: Python** — Chosen for its robust data handling and automation capabilities.
- **Integrated Development Environment (IDE): VS Code** — Leveraged for script development, real-time terminal debugging, and environment management.
- **Core Libraries:**
    - **Pandas:** Primary engine for DataFrame manipulation, structural auditing, and handling 181 distinct table sheets.
    - **xlrd & openpyxl:** Specialized engines implemented to ensure full compatibility.
- **Data Transformation Logic:**
    - **Header-Bypass Logic:** Implemented ‘skiprows=4’ to programmatically isolate the clinical statistical matrices from decorative publication headers.

- **Null-Handling:** Utilized ‘dropna()’ to distinguish between Structural Nulls (formatting white space) and Suppressed Data (clinical safety protocols).

3. Dataset Inventory & Scope:

The audit covered 13 specialized domains, ensuring a holistic view of the UK’s mental health landscape.

Domain	Key Chapters Audited	Table Count	Technical Utility
General Prevalence	Ch 2, 2024 Update	~25	Baseline for CMD risk modeling.
Clinical Conditions	Ch 4–9 (PTSD, ADHD, etc.)	~90	Feature engineering for specific disorders.
Crisis & Behaviors	Ch 10–13 (Suicide, Drugs)	~50	Anomaly detection & crisis intervention logic.
Meta-Data	Ch 14 & Specifications	~16	Weighting variables & sampling design.

152	apms-2014-ch-12-tabs.xls	12.10	18	4	38
153	apms-2014-ch-12-tabs.xls	12.11	16	4	32
154	apms-2014-ch-12-tabs.xls	12.12	16	4	32
155	apms-2014-ch-12-tabs.xls	12.13	14	4	32
156	apms-2014-ch-12-tabs.xls	12.14	14	4	32
157	apms-2014-ch-12-tabs.xls	Table specification	13	2	0
158	apms-2014-ch-13-tabs.xls	13.1	40	6	86
159	apms-2014-ch-13-tabs.xls	13.2	41	8	136
160	apms-2014-ch-13-tabs.xls	13.3	27	11	85
161	apms-2014-ch-13-tabs.xls	13.4	49	8	129
162	apms-2014-ch-13-tabs.xls	13.5	51	7	91
163	apms-2014-ch-13-tabs.xls	Table specification	4	2	0
164	apms-2014-ch-14-tabs.xls	14.1	14	4	22
165	apms-2014-ch-14-tabs.xls	14.2	18	5	33
166	apms-2014-ch-14-tabs.xls	14.3	15	3	22
167	apms-2014-ch-14-tabs.xls	14.4	16	2	13
168	apms-2014-ch-14-tabs.xls	14.5	31	5	42
169	apms-2014-ch-14-tabs.xls	14.6	39	9	133
170	apms-2014-ch-14-tabs.xls	14.7	19	9	85
171	apms-2014-ch-14-tabs.xls	14.8	10	9	53
172	apms-2014-ch-14-tabs.xls	14.9	7	9	51
173	apms-2014-ch-14-tabs.xls	14.10	7	9	51
174	apms-2014-ch-14-tabs.xls	14.11	15	9	74
175	apms-2014-ch-14-tabs.xls	14.12	7	9	51
176	apms-2014-ch-14-tabs.xls	14.13	7	9	51
177	apms-2014-ch-14-tabs.xls	14.14	16	9	83
178	apms-2014-ch-14-tabs.xls	14.15	18	9	91
179	apms-2014-ch-14-tabs.xls	14.16	34	9	149
180	apms-2014-ch-14-tabs.xls	Table specification	10	2	0

```
[181 rows x 5 columns]
```

4. Technical Profiling Insights:

- **Structural Layout:** The files use a Hierarchical Matrix format where sheet names (e.g., "2.1") act as primary keys for specific demographic intersections.
- **Variable Density:** Tables vary from 4 to 34 columns. The high-density tables (30+ cols) provide the multi-variate correlations (e.g., Condition vs. Employment vs. Region) necessary for deep learning.
- **Data Integrity:** Missing values are identified as Structural Nulls (formatting) or Suppressed Data (to protect anonymity in small sample sizes), ensuring we do not misinterpret zero-values as errors.

All adults — *2014*

Clinical Interview Schedule – Revised (CIS-R) score[a]	Age							
	16-24	25-34	35-44	45-54	55-64	65-74	75+	All
	%	%	%	%	%	%	%	%
Men								
0-5	76.5	72.3	69.7	72.1	69.9	81.3	81.9	74.0
6-11	14.4	12.5	15.2	14.7	15.2	11.5	12.8	13.9
Under 12	**90.9**	**84.7**	**84.9**	**86.8**	**85.1**	**92.7**	**94.7**	**87.8**
12-17	4.9	7.4	6.8	6.2	5.8	3.7	4.2	5.8
18 or more	4.2	7.8	8.3	7.0	9.1	3.6	1.1	6.4
12 or more	**9.1**	**15.3**	**15.1**	**13.2**	**14.9**	**7.3**	**5.3**	**12.2**
Women								
0-5	57.3	60.4	63.2	58.6	62.8	72.3	74.6	63.5
6-11	16.7	20.5	16.2	18.7	18.1	14.9	15.4	17.4
Under 12	**74.0**	**80.9**	**79.4**	**77.3**	**80.9**	**87.1**	**90.0**	**80.9**
12-17	10.9	10.5	9.4	10.2	9.8	8.1	5.1	9.3
18 or more	15.1	8.6	11.2	12.5	9.3	4.8	4.9	9.8
12 or more	**26.0**	**19.1**	**20.6**	**22.7**	**19.1**	**12.9**	**10.0**	**19.1**
All adults								
0-5	67.1	66.3	66.4	65.3	66.3	76.6	77.6	68.6
6-11	15.5	16.5	15.7	16.7	16.7	13.2	14.3	15.7
Under 12	**82.7**	**82.8**	**82.1**	**82.0**	**82.9**	**89.8**	**91.9**	**84.3**
12-17	7.8	9.0	8.1	8.2	7.8	5.9	4.7	7.6
18 or more	9.5	8.2	9.8	9.8	9.2	4.2	3.3	8.1
12 or more	**17.3**	**17.2**	**17.9**	**18.0**	**17.1**	**10.2**	**8.1**	**15.7**
Bases[b]								
Men	*249*	*355*	*468*	*489*	*541*	*538*	*418*	*3,058*
Women	*311*	*680*	*712*	*805*	*685*	*651*	*644*	*4,488*
All	*560*	*1,035*	*1,180*	*1,294*	*1,226*	*1,189*	*1,062*	*7,546*

```
--- CLEAN DATAFRAME PREVIEW (TABLE 2.1) ---
   Unnamed: 0     %   %.1   %.2   %.3   %.4   %.5   %.6   %.7
0         Men   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN
1         0-5  76.5  72.3  69.7  72.1  69.9  81.3  81.9  74.0
2        6-11  14.4  12.5  15.2  14.7  15.2  11.5  12.8  13.9
3    Under 12  90.9  84.7  84.9  86.8  85.1  92.7  94.7  87.8
4       12-17   4.9   7.4   6.8   6.2   5.8   3.7   4.2   5.8
5  18 or more   4.2   7.8   8.3   7.0   9.1   3.6   1.1   6.4
6  12 or more   9.1  15.3  15.1  13.2  14.9   7.3   5.3  12.2
7       Women   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN
8         0-5  57.3  60.4  63.2  58.6  62.8  72.3  74.6  63.5
9        6-11  16.7  20.5  16.2  18.7  18.1  14.9  15.4  17.4
PS C:\Users\priya\OneDrive\Desktop\AI MENTAL HEALTH ASSISTANT>
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5. Conclusion

The APMS profiling workflow successfully established a reliable analytical foundation for downstream healthcare analytics and mental health trend analysis. Structural auditing confirmed that the dataset is statistically robust, methodologically documented, and suitable for descriptive analytics, prevalence analysis, and predictive modelling preparation. The profiling process also highlighted the importance of contextual interpretation of structural missingness and publication-oriented spreadsheet layouts commonly observed in NHS statistical datasets.